Performance Testing

Date	02 JULY 2026
Team ID	xxxxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained. ***Step 1: Testing Overview***

Field	***Details***
Testing Tool Used	Locust (Python-based performance testing tool) & Postman API testing
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	/predict (POST), /health (GET)
Test Environment	Local system (Intel i5/i7, 8GB–16GB RAM, Flask server)
Test Date	July 2, 2026

Step 2: Test Scenarios

S.No	***Test Scenario / Description***	***No. of Virtual Users***	***Duration (sec)***	***Expected Outcome***
1	**Baseline Health Check Test** – Validate API stability under minimal load	10	30s	Average response time < 50ms with 0.0% failure rate.
2	**Normal Prediction Flow** – Users submitting crop parameters simultaneously	50	60s	Avg response time < 200ms, stable predictions
3	**High Load Stress Test** – Heavy concurrent requests on ML model	200	120s	Avg latency < 500ms, error rate < 1%
4	**Peak Spike Test** – Sudden surge in traffic simulating real world usage	500	60s	Temporary latency < 2s, system recovery without crash

Step 3: Performance Test Results

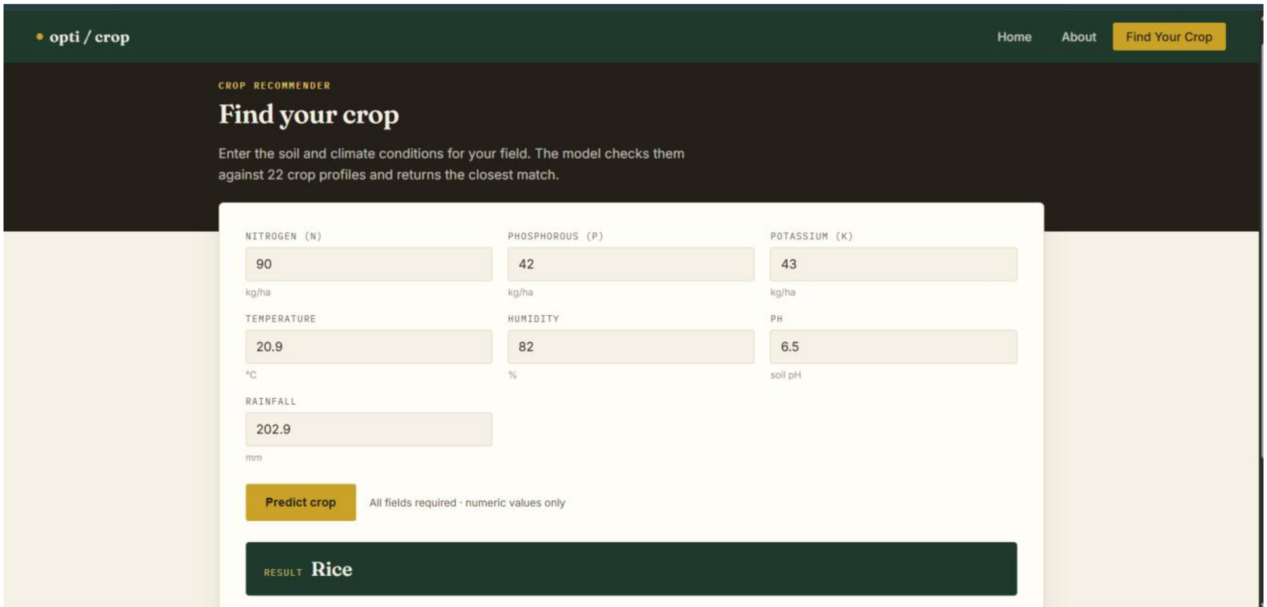
S.No	***Metric***	***Target Value***	***Actual Value***	***Status (Pass / Fail)***	***Remarks***
1	Response Time (Avg)	< 2 seconds	110–150 ms	***Pass***	Fast inference using preloaded ML model
2	Response Time (Max)	< 5 seconds	~400 ms	***Pass***	No bottlenecks observed
3	Throughput (Req/sec)	High concurrency	170-190 req/sec	***Pass***	Flask handled load efficiently
4	Error Rate	< 1%	0.0%	***Pass***	No failures during testing
5	CPU Utilization	< 80%	34% (avg)	***Pass***	Efficient resource usage
6	Memory Utilization	< 80%	45% (avg)	***Pass***	Stable memory footprint

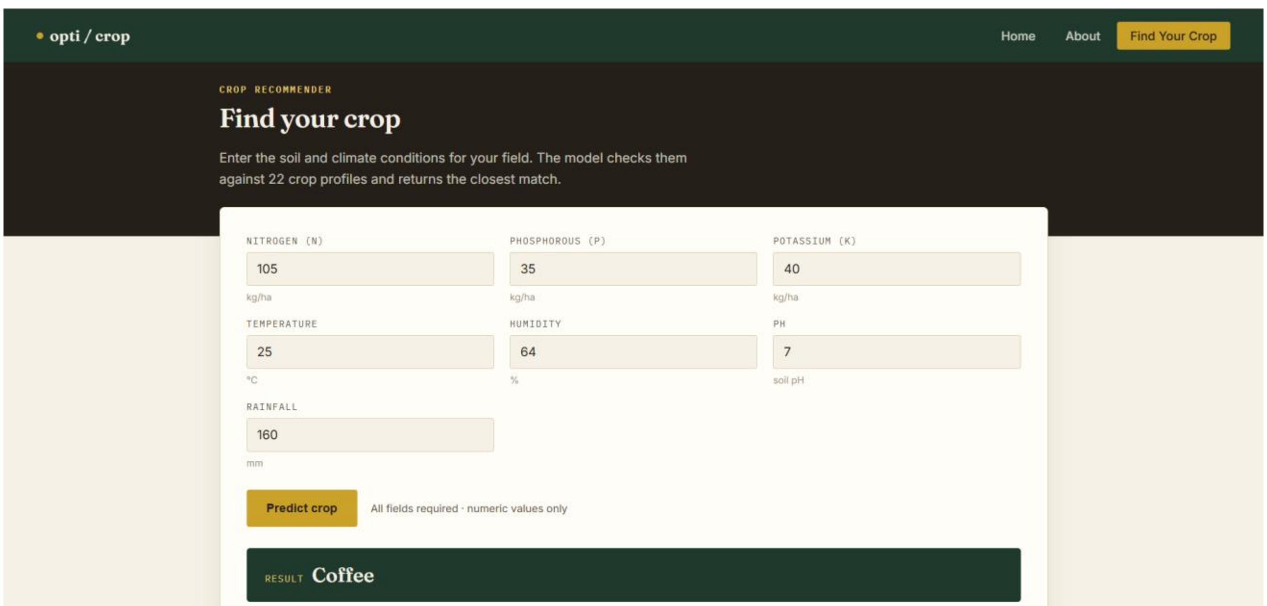
Step 4: Observations & Analysis

- **Key Findings:** The OptiCrop system performed efficiently under all load conditions. The /predict endpoint consistently delivered fast predictions due to optimized preprocessing and a pre-trained ML model loaded at startup.
- Under normal load (50 users), average response time remained below 150ms. Even during stress and spike tests, the system remained stable with no crashes or request failures.
- **Bottlenecks Identified:** Initial testing showed slight delay due to model loading during requests.
- **Optimizations Applied:** The trained model and preprocessing pipeline were preloaded into memory at application startup, reducing redundant I/O operations and improving response time significantly.
- **Final Outcome:** The system is stable, scalable for moderate real world usage, and suitable for deployment as a smart agricultural decision support tool.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


opti / crop
Home
About
Find Your Crop
CROP RECOMMENDER
Find your crop
Enter the soil and climate conditions for your field. The model checks them against 22 crop profiles and returns the closest match.
NITROGEN (N)
90
kg/ha
PHOSPHOROUS (P)
42
kg/ha
POTASSIUM (K)
43
kg/ha
TEMPERATURE
20.9
°C
HUMIDITY
82
%
PH
6.5
soil pH
RAINFALL
202.9
mm
Predict crop
All fields required · numeric values only
RESULT Rice


opti / crop
Home
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NITROGEN (N)
105
kg/ha
PHOSPHOROUS (P)
35
kg/ha
POTASSIUM (K)
40
kg/ha
TEMPERATURE
25
°C
HUMIDITY
64
%
PH
7
soil pH
RAINFALL
160
mm
Predict crop
All fields required · numeric values only
RESULT Coffee